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Replicability and scalability of mini-grid solution to rural electrification programs in sub-Saharan Africa

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ABSTRACT

The assessment of off-grid electrification programs in developing countries largely based on mini-grid and solar home system (SHS) has shown that they are faced with low development imparts and sustainability challenges, which has resulted in failure of many projects. This study provides solutions on how to surmount these challenges, leaning on the experience of a hybrid solar-diesel mini-grid at Tsumkwe village in Namibia. It provides analyses of a case study based on empirical evidence from field studies, interviews of representatives of households, public institutions and energy providers. In addition, it investigates the technical challenges and economic impacts of the electrification program. HOMER™ and MATLAB™ models were used in the analysis and investigations. The findings show that despite the challenges, the system has been sustained because it keyed into an existing structure with growth potentials. The progressive tariff system adopted by the government helped to cushion costs and allow low income households in the energy matrix. Adoption of strict maintenance measures, and implementation of energy efficiency measures prior to the commissioning of the program, resulted in the reduction of costs. The success elements identified in this study could be extrapolated in other sub-Saharan African countries if the challenges are properly addressed.

1. Introduction

The world energy trend shows that developing nations remain an outlier in terms of access to modern energy services. Previous reports indicate that more than 1.3 billion people have no access to electricity globally and more than 95% of these people live in either sub-Saharan Africa or developing Asia, worst still 84% of them live in rural areas [1,2]. Projections show that with the current business-as-usual approach and the bourgeoning population this trend is unlikely to improve significantly in future [3]. Therefore, increasing electricity access in developing countries remains a priority interest and major developmental objectives of most governments. Traditionally, increasing electricity access to previously un-electrified areas were done using grid expansion programs. The centralized grid approach ignored limitations and peculiarities of the remote rural and island locations [4]. This situation makes it difficult to increase electricity access in remote rural areas and Island settlements due to lack of patronage by energy providers. However, with the evolvement of technology, enabling policy frameworks and robust financing models, successful off-grid electrification programs have been implemented in many countries using technologies such as mini-grid and solar home system (SHS). Furthermore, increasing need to reduce carbon emissions in fossil base grids, increasing cost of grid expansion, reducing costs of renewable energy alternative and increasing performance, and inaccessibility of various settlements also favoured the distributed generation approach [5]. However, sustainability of operation of SHS are constrained due to the limited power capacity and lack of development impacts [6–8]. Therefore, solution to rural electrification is tilting more in favour of the mini-grid alternative by the day. In the global arena, mini-grid is used in the developed world to boost energy security, power quality and reliability, as well as to forestall blackouts during contingencies [5]. This however is not the case in the developing world, mini-grids are used to bring about power availability through increasing access to electricity mostly in the remote rural and island settlements. Nevertheless, mini-grids are necessary in island communities whether in the developed or developing countries.

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Despite challenges, countries in Latin America such as Costa Rica Brazil, Mexico, Argentina and Chile and developing Asia such as Thailand, Nepal, India Sri Lanka, Indonesia, Cambodia, Bangladesh and China have recorded success in rural electrification programs using mini-grid [9–12] and [4].

On the other hand, numerous publications have reported the penetration of mini-grid in African countries such as, Nigeria, Mali, Senegal, Mauritania, Cameroon, Kenya, Uganda, Rwanda, Tanzania, Algeria, Madagascar Lesotho and South Africa [13–17]. Also, rural electrification programs of Morocco were largely achieved through mini-grid [18].

In spite of the successes recorded in other developing countries, the sub-Saharan African experience is a narrative of failure and limited success. Implementation of off-grid electrification programs have been fraught with challenges, resulting in high mortality rate of projects [8,19]. Some of the constraints militating against the sustainability of renewable energy electrification projects include challenges such as lack of technical knowledge needed to run and maintain the systems, disperse homesteads, low energy demand density, none availability of spare parts, inappropriate financing models, lack of operational and managerial skills required to run and sustain projects after commissioning. In Zimbabwe a GEF- UNDP funded off-grid electrification program did not survive after the project came to an end [19]. A hydro base mini-grid systems implemented in Lesotho had limited success due to maintenance issues and high production cost [17]. The off-take of a hybrid solar-biogas mini-grid project at Sekhutlane village in Botswana was suspended due to technical and managerial challenges [20]. Lucingweni pilot hybrid mini-grid project in South Africa did not survive beyond three months of operation after commissioning due to an inadequate feasibility study, high electricity production cost, insufficient community engagement, and lack of expertise required to run and manage the plant [21,22]. Also project failures has been attributed to policy fragmentations in developing countries [23]. Another reports argued that even though there is high potential for the establishment of low carbon hybrid mini-grid in a country like Nigeria, policy inconsistency and lack of political will has hindered the off take of projects [24,25].

Empirical evidence in the world development trends indicates that wealthy and developed nations have better electricity access than poor and developing nations. The fact that electricity access and development are inextricably linked underscores the need for provision of sustainable energy access in the developing world. A previous report argued that electrification cannot solve all the development problems of the rural households, but these households cannot avail themselves of development assistance opportunities without access to electricity [26]. Despite high level talks among mainstream institutions such as the World Bank (WB), United Nations Development Program (UNDP), Global Environment Facility (GEF) and governments supporting rural electrification programs in developing countries, the socio-economic development impacts of the programs have remained inconsequential [7,8]. The available reports show that there is limited information on the field performances of the technologies used for rural electrification programs in developing countries. Investigations of the field performance of several rural electrification programs show that the sustainability of most programs faces major challenges [27–29].

Despite the unsuccessful mini-grid projects elsewhere in Southern Africa, Tsumkwe hybrid solar-diesel mini-grid in Namibia, one of the largest systems in sub-Saharan Africa, has been relatively successful, having been in operation since its commissioning in 2012.

To identify the factors that influence the success of mini-grid systems in rural areas of southern Africa, this study presents an assessment of a solar-diesel hybrid mini-grid in Tsumkwe, Namibia. It addressed the socio-economic and development impacts of the mini-grid electrification program, the technical dimensions and the challenges that could affect sustainability of the system.

The objective of this study is to identify success factors and challenges from a functioning mini-grid installation in southern Africa. The study sets out to answer the following questions:

- What are the success factors of the Tsumkwe hybrid mini-grid?
- What are the challenges to the sustainability of the Tsumkwe hybrid mini-grid?
- How can the lessons learned from the Tsumkwe hybrid mini-grid contribute to rural electrification of off-grid communities in sub-Saharan Africa?

2. Study area

Tsumkwe village is situated in Otjozondjupa region in the north-eastern part of Namibia, about 300 km east of Grootfontein, which is the nearest town, and about 54 km to the border with Botswana (Fig. 1). According to Namibia Population and Housing Census (NPC) of 2001, Tsumkwe constituency had about 9000 inhabitants, of which approximately 3800 lived in Tsumkwe village [30]. The current population of Tsumkwe is not known since there has been an influx of people from neighboring villages following improved access to electricity after the commissioning of the hybrid solar-diesel mini-grid.

The people of Tsumkwe are predominantly San of the Ju|'hoansi tribe but the residents come from all over Namibia, many with their own language and culture. The most commonly spoken language is Afrikaans, but there is a large number of people who only speak Ju|'hoan, the local San language. There are also Damara, Oshiwambo, Caprivi, Kavango and Otjiherero speaking minorities who originally were not natives. Traditionally San rely on nomadic hunter-gathering for subsistence and their income generating capacity is generally low [31]. Even though there are several government ministries, non-governmental organizations and small and medium sized businesses in the village, there is limited employment potential for a majority of the population [32].

Tsumkwe is located far from the national grid. According to [31],

Fig. 1. The location of Tsumkwe in Namibia.

the nearest power lines to Tsumkwe is a 33 kV line about 180 km away, and a 66 kV line about 240 km away. Due to the long distance from the grid and the high level of reactive power required for long distance transmission, these two alternatives are not suitable. A 132 kV line is required to transmit power over such a long distance, and the nearest one is located at Tsumeb distribution station more than 300 km away [33]. It has been estimated that it would cost about US$ 15 million (about N$ 150 million at a foreign exchange rate of N$10 to US$1) to extend the grid from Tsumeb station to Tsumkwe, which is regarded as too much given the expected low electricity consumption of the inhabitants [30]. Prior to the installation of the hybrid solar-diesel system in Tsumkwe, there was a mini-grid in place that was powered by two diesel generators, with capacities of 350 kVA and 285 kVA, providing parts of the village with electricity [30,33]. However, due to high electricity production cost, electricity tariff was so high that a majority of the households could not afford the cost. Also, the use of credit meters gave room to accumulation of debts. In spite of this, the power supply was unreliable due to the poorly managed diesel generators, high cost of diesel and poor logistics in the supply, which limits the generator operations to between 10 and 14 h per day, for this reason, sometimes most users are left without electricity for days [30].

In order to develop a more sustainable electrification program for the rural settlements in Namibia, Tsumkwe hybrid mini-grid was designed to supply electricity for domestic, commercial and public activities, with the intention of improving income generating capacity of the users. In March 2008 the European Union (EU) funded proof of concept project was initiated by the Desert Research Foundation of Namibia (DRFN) in collaboration with the European Commission (EC), NamPower and the Otjozondjupa Regional Council (OTRC). The budget for the implementation of the hybrid mini-grid was about US$3 million (N$30.8) [30]. At the time of commissioning the Tsumkwe diesel-solar mini grid was the largest hybrid mini-grid system in southern Africa (Fig. 2).

The hybrid system consists of 918 polycrystalline solar panels with a total rated power of 202 kWp, the battery bank consists of 766 kWh lead acid deep cycle battery storage capacity, while the generator set comprises of two 150 kVA and one standby 350 kVA diesel generators. The system is designed to provide a 24-h supply of electricity to Tsumkwe village [30]. The program was commissioned in 2012 and still operational to date despite facing several challenges.

3. Methods

The data for this study were collected through observations, empirical evidence, sampling and semi structured interviews of a majority of the stakeholders in Tsumkwe energy program. Semi structured interview method was adopted to enable the interviewees express their opinions independently. Also information from the plant managers, and logged data from the mini-grid system form the basis for the techno-economic analysis in the study. Furthermore, the study also utilizes information gathered from project documents in the analysis. The data gathering was conducted by a team of researchers from Malardalen University Sweden, assisted by Tsumkwe energy project implementation coordinator who speaks the local language and consequently introduced us to the locals to gain their confidence before the interviews.

3.1. Social and economic impacts of the hybrid solar-diesel mini-grid electricity

The paper sets out to analyze qualitatively the social and economic impacts of Tsumkwe energy program. Based on this, 20 selected key informants who had experience of the electricity situation before and after the introduction of the solar-diesel hybrid mini-grid in Tsumkwe village were interviewed. They ranged from technical staff operating and managing the plant, administrative staff of Otjozondjupa Regional Council (OTRC) who are the custodians of the system, small and medium scale enterprises to public institutions, and individual consumers represented by household heads (Table 1).

3.2. Techno-economic analysis of the hybrid solar-diesel mini-grid

The analysis of the renewable energy systems price trends and load profiles of the hybrid mini-grid system is based on MATLAB™ model. The techno-economic viability of the hybrid mini-grids was done using financial instruments such as levelized cost of electricity (LCOE), net present cost (NPC), Initial Capital Cost (ICC), and operating cost (OC). HOMER™ hybrid energy optimization model was used to obtain the energy outputs and economic indicators. Simulations were performed based on system costs obtained from project documents. The results obtained were then compared with the hourly logged data. HOMER model utilizes equation (1) in calculating the power production capacity from the PV and the diesel generator systems.

$$P(t) = \sum_{S=1}^{N_S} PV_S + \sum_{W=1}^{N_G} P_G \tag{1}$$

where, PV_S is the electrical power output from the solar PV array and P_G is the generator power output. The total electrical power output from the PV_S is given by equation (2)

$$PV_S = Y_{PV} f_{PV} \left(\frac{G_T}{G, STC}\right) [1 + \alpha_P (T_C - T_{C,STC})] \tag{2}$$

where f_{PV} is PV derating factor [%], Y_{PV} is PV rated capacity (kW), G_T is the incident global irradiation (kW/m^2), G, STC is the incident radiation at standard test conditions (1 kW/m^2), α_P is temperature coefficient of power (%, °C), T_C is PV cell temperature (°C) and $T_{C,STC}$ is PV cell temperature under standard conditions (25 °C).

Generator (diesel) total electrical power output is given by equation (3)

$$P_G = \frac{F - F_0 Y_G}{F_1} \tag{3}$$

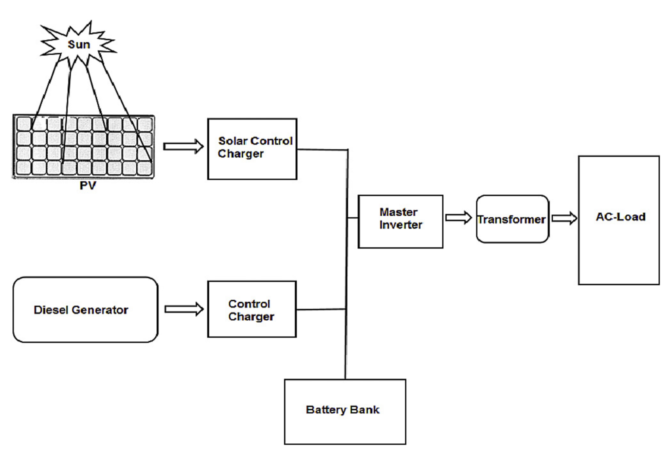

Fig. 2. Layout of the PV-diesel hybrid mini-grid in Tsumkwe, Namibia.

Table 1
Key informants interviewed during the field survey in Tsumkwe, Namibia.

Respondents	Government users	Domestic, Commercial and Institutional users	Designations
Households	Tsumkwe police staff quarters	San, Herero, Owambo, Kavango, Damara-Nama, and Caprivi dwellings	Household Heads
Energy provider	Otjozondujupa Regional Council (OTRC), plant manager & attendants		Head (Department of Works and Chief Admin Officer (OTRC), Technician (plant), engineers (NamPower)
SME	Tsumkwe NamWater pump	Retail shops, general dealer, hospitality business, Tsumkwe craft center	Entrepreneurs and business managers
Government	Community development center	Nyae Nyae Conservancy, Namibian Broadcasting Corporation	Program Officer and manager
Health	Clinic		Head Nurse
Schools	Tsumkwe Secondary School		Principal
Security	Police		Tsumkwe Unit Commander, criminal investigation department (CID)

where, P_G is the electrical output of the generator, F is the fuel consumption rate (l/h), F_0 is the fuel curve intercept coefficient ($L/h/kW_{rated}$), Y_G is the rated capacity of the generator (kW), and F_1 is the fuel curve slope ($L/h/kW_{output}$).

The load profile of the hybrid mini-grid in Tsumkwe was computed from data recorded hourly by the master inverter (Fig. 2) from January to December in 2014.

3.3. Limitations and assumptions of the study

It was assumed that the foreign exchange (FOREX) rate for Namibian Dollar (NAD) and USA Dollar (US$ or $) was constant at US$1.00 to N$10.00. The techno-economic analysis based on HOMER™ model did not take government subsidies into account. Due to time constraints and resources, we were not able to cover all respondents in the study as intended. Information from project document shows a lot of discrepancies on the electricity tariff for diesel only mini-grid prior to the implementation of the solar-diesel hybrid mini-grid, even though they all suggest it was high. Also, there were inconsistencies on the actual prices of equipment and materials used during the implementation of the project, therefore our costs were based on averages. The presence of the project coordinator could have influenced the answers given by respondents during the interview. However, responses were counter balanced by the empirical evidence observed during the interview. For the economic analysis, the costs of the PV array and generator were based on the cost at the time of construction. Therefore, the results may not accurately represent the current situation due to changes in interest rates, inflation, price of diesel and other variables that have occurred since the initial investment. Nevertheless, the analysis provides valuable information about the technical, economic and social challenges that affect the sustainability of the solar-diesel hybrid mini-grid in Tsumkwe village.

4. Results

4.1. Socio-economic dimensions of Tsumkwe hybrid mini-grid

This section presents the tariff system adopted for the Tsumkwe mini-grid system followed by an analysis of the social and economic impacts of the hybrid mini-grid program on the inhabitants of Tsumkwe village. The perceived impacts of the mini-grid on the households' livelihoods as well how it has affected businesses and other income generating activities were analyzed. This section also considers the challenges facing the sustainability of the hybrid mini-grid and factors contributing to its success.

4.1.1. Tsumkwe mini-grid tariff system

The success of Tsumkwe mini-grid can be attributed to the sound financial management scheme adopted by OTRC, who own and maintain the system. An efficient revenue collection method is implemented via prepaid metering of the electricity supply to users. Revenue is managed through the establishment of an escrow account to ensure that money collected is reinvested in the system for maintenance, replacement and upgrade of the batteries, generators, PV arrays, power transmission, distribution equipment and other paraphernalia of power environment products.

A dual tariff system that takes into consideration the residential, commercial and institutional users is applied for service charges. The tariff for residential users is US$0.15/kWh and for commercial and public institutions it is US$ 0.25/kWh as of 2015. The idea is to use the revenue from institutional and high energy consuming commercial users to balance that from the low income users, and to ensure that different demographic groups are able to afford the electricity provided. This has helped to some extent, even though the revenue does not currently meet the levelized cost of electricity. The disparity is bridged by government subsidies. According to the program the subsidy will be in place for nine years after commissioning, i.e. until 2021, when it is estimated that Tsumkwe mini-grid system will gain financial independence. An average household in Tsumkwe spends about US$15 per month on electricity. There is high disparity in how much the business customer pay, depending on the size of the business and the energy demand. For instance, Tsumkwe Lodge, one of the biggest commercial users in Tsumkwe, pays US$ 600 for electricity per month during low season and about US$ 1200 per month during high season. The grocery stores also known as “general dealer” pays US$ 720 for electricity per month. However, respondents from small and medium scale enterprise (SME) indicate that they pay US$15-30 per month. The connection fee for domestic connection is US$ 5, while for commercial and public institutions is US$ 32. Despite a few cases of electricity theft, the willingness to pay for services is very high in Tsumkwe.

4.1.2. Social impacts of Tsumkwe hybrid mini-grid system

According to interviewees the provision of street lights along major streets has improved safety in the village at night. Several of the interviewed domestic users stated that electricity has improved their standard of living as they no longer have to use firewood to keep warm during winter. Students can now extend their study time to after sunset. Households with children in school stated that their children perform better in school since they now have access to electric light.

The population in Tsumkwe has increased since the commissioning of the hybrid system. According to the regional officials in Tsumkwe, this is evidenced by the increased number of applications for land plots and increased number of buildings and businesses in the village. Several ministries have opened offices in the village after the access to electricity was improved. The increased population has resulted in a higher load on the system according to the plant operators, more electrical appliances are being connected to the mini-grid. According to the information, more reliable electricity supply to a larger part of the population has contributed to an improved standard of living in Tsumkwe, which is evident from the increasing number of vehicles delivering refrigerators and other appliances to households. A major advantage of 24-h electricity is the increased storage capacity for food in refrigerators. One respondent stated that households eat more fresh foods than before, since perishable foodstuff can now be stored.

Business owners stated that better access to electricity has improved their business. Several new businesses have been established. In the past the electricity costs for businesses were higher as they had to operate their own diesel generators for the time when there was no electricity provided from the diesel only mini-grid. This contributed to increased overhead costs of businesses and to air and noise pollution.

The establishment of Tsumkwe Community Development Center (TCDC) has provided a platform for interested and willing individuals to acquire computer skills and also facilitates easy communication with the outside world through the provision of computers and allied services.

According to representatives from the police force in Tsumkwe, the improved access to electricity and electric lighting has had a positive effect on crime prevention and aided arrests due to increased visibility. Nevertheless, the number of assaults, affray and rape has increased due to the increase in active nightlife, while the number of cases of theft has decreased. The 24-h electricity supply has improved the storage of corpses in the police mortuary, which was a major problem in the past.

Respondents from the clinic stated that health care delivery has improved because of the improved access to electricity. In the past they used candles and torches when delivering babies at night and storage of vaccines was not possible. Now they can use sophisticated medical equipment and store vaccines in refrigerators. Corpses can now be stored in the morgue until burial without problems.

According to the principal of the secondary school in Tsumkwe, the performance of students has not shown significant change since the access to electricity was improved. Nonetheless, the presence of electricity has enhanced the learning programs and the administration, allowing for computers to be used in classes, by teachers and by administrative staff. The hostel now has warm water unlike in the past. Electrical appliances such as deep freezers, cooler rooms, electric pots, computers, printers, photocopiers, scanners and projectors are now used at school.

4.1.3. The economic impacts of Tsumkwe hybrid mini-grid

Responses from the interviewees suggest that the improved access to electricity is motivating people to venture into businesses or expand existing ones. According to the manager of the grocery store, they expanded their business after the building of the hybrid mini-grid by providing new services like supply of LPG, 24-h access to the fuel station with petrol and diesel, and an ATM machine.

According to the respondents the number of bars and other drinking places has increased dramatically since the introduction of the 24-h electricity supply, which has had both positive and negative effects on the community. On one hand the bars are contributing to the local economy, but they also lead to more alcohol abuse, fighting and crime, which have been noted by the local police.

4.1.4. Challenges to the sustainability of the Tsumkwe hybrid mini-grid

According to the technical staff in charge of operation and maintenance of the hybrid mini-grid system, the major challenge affecting the operation of the plant is the supply of spare parts. Many components of the hybrid system have to be imported from overseas. Another problem is that the cooling system for the generators is not optimal, which results in dust frequently blocking the radiators, causing overheating and unscheduled interruptions of the electricity supply.

The frequency of fuse breakages due to overload has increased. This is believed to be an effect of increasing energy demand and use of non-optimal appliances. Related to the increasing demand of electricity and the use of less efficient equipment, a majority of the representatives from households, businesses and public institutions stated that they experience fluctuation in the electricity supply, which has damaged both household and business appliances. Fluctuations in the electricity supply are also caused by the issues of the synchronization of the generators with the PV system, resulting in manual operation and unscheduled interruption during the process.

Another challenge to the users of electricity from the hybrid system is that sales of electricity credit token is restricted to only one vendor (the OTRC administrative office), and whenever the office is closed anyone in need of token has to wait until it reopens. Buying of token is therefore a major challenge for users whenever the vendor is not around, resulting in businesses and domestic activities being stalled.

Even though the Tsumkwe mini-grid is currently managed by dedicated staff, the human resource situation is challenging as the present head of operations is due to retire. There is currently no one else in the village with the technical skills required to run and maintain the system.

4.2. Technical dimensions of Tsumkwe hybrid mini-grid

This section presents the results of the technical analysis of Tsumkwe mini-grid. It gives detail information on the load profile, seasonal load variation, load increase from 2012 to 2015, status of operation of the mini-grid, energy production capacity and the economics of operation of the solar-diesel hybrid mini-grid.

4.2.1. Load profile of Tsumkwe village

Information from project documents show that four energy efficiency measures were adopted prior to the commissioning of the project. This involves conversion of existing incandescent bulbs to compact fluorescent lamps (CFL), conversion of streetlights from mercury vapour bulbs to high pressure sodium bulbs, provision of solar water heater (geyser) in place of electric water heater, provision of gas stoves using liquefied petroleum gas (LPG) instead of electric stoves and hot plates. These measures helped in reducing the load by about 34% [34].

The electricity supply network is divided into two categories comprising the essential and non-essential lines. The essential line provides electricity to priority users, i.e. the clinic, water pumps, police station and two schools. The non-essential line provides electricity to domestic and commercial users. Electricity provision is prioritized for the essential clients. The analysis of the load data shows that on the average, electricity consumption in Tsumkwe village is about 1763 kWh/day. The consumption on the non-essential line is about 1586 kWh/day on the average. The remaining 12% is used by the essential clients. The load profile indicates that Tsumkwe has baseload at night and peak load during the day and evening due to increased commercial and domestic activities (Fig. 3). The relative high electricity demand during the night is

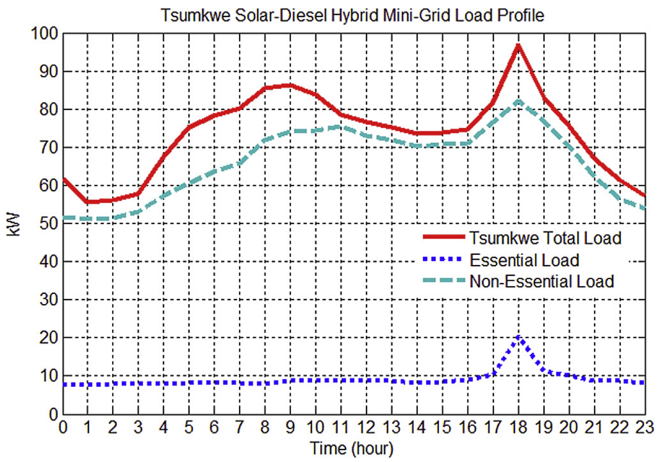

Fig. 3. Daily electricity consumption profile in Tsumkwe based on 2014 data.

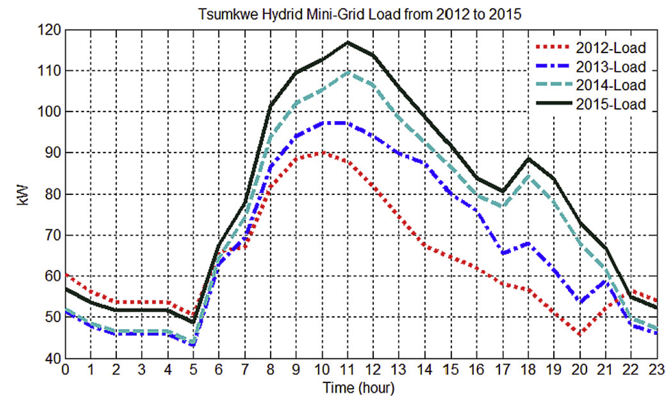

Fig. 5. Load profile of Tsumkwe mini-grid from 2012 to 2015.

attributable to street lights and increased commercial activities at night as a result of extended hours of business by shops and bars (known locally as Shebeen).

4.2.2. Seasonal load variation

The analysis of the seasonal load shows that there is a significant increase in the load during summer in contrast with winter period (Fig. 4). The usage pattern indicates that the load increases by about 26% during summer. This is mainly due to the increased nightlife with the attendant increase in the use of inductive loads such as refrigerators, air conditioners and fans, also restive loads like water heaters and electric stoves has been reported to be used by some nonconformist that refused to adhere to the energy efficiency measures.

4.2.3. Load trend from 2012 to 2015

Information from the logged data from year 2012–2015 indicates that the load demand is on the increase (Fig. 5). The load profile for 2012 like the other years had the peak loads during the day and baseload at night, however the night load is disproportionately higher. This period marked the transition to energy efficiency measure adopted before the commissioning of the mini-grid. There is an increase in the load between year 2012 and 2013, the load increased by about 5%. This could be attributed to new connections from low income residents who previously lacked access to electricity. Between 2013 and 2014 the load increased sharply due to the sudden increase in population as well as increased domestic appliances, resulting from the awareness of the hybrid mini-grid in Tsumkwe constituency. This trend continued to 2015 with an increase of 7%, it is expected that the load will continue to increase with population growth, and as income increases, more household appliances would be added to the mini-grid.

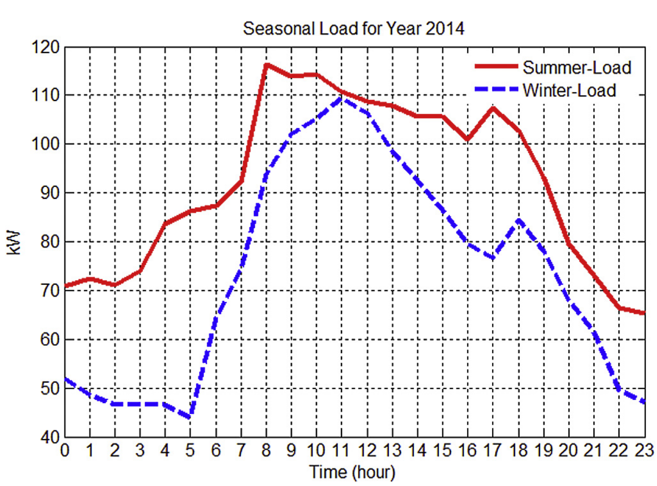

Fig. 4. Differences in summer and winter loads in 2014.

4.2.4. Status of operation of the hybrid mini-grid in Tsumkwe

During the day electricity is supplied from the PV array. Additional electricity can be supplied from the battery bank and from the generators if the demand is higher than the array can deliver (according to the plant operator). Under normal conditions the generator starts up at 6 p.m. and runs for about 3 or more hours, providing electricity to the village and charging the batteries. When the batteries are charged and the generator is switched off, the battery bank supplies the village with electricity until the early morning hours, when the PV array starts to produce again.

The batteries are operated at a state of charge (SOC) of between 55% and 95% to maintain the depth of discharge (DOD) at a safe level. Analysis of logged data shows that the batteries are constantly charging when the generator is ON, but alternate between charging and discharging states whenever electricity is being supplied by the PV array. On clear days the generator operates according to the above schedule, before the battery takes over the supply until the SOC reaches about 55%. The supply then reverts either to the generator or the PV array depending on the time of day. On overcast days, which are most common during the rainy season in summer, the generator operation is less regular, resulting in more frequent switching ON and OFF operation to make up for the shortfall from the PV array. However, due to the increased solar irradiation during the summer period, the battery system is consistently on charge mode during the day. The expected life span of the lead acid deep cycle battery is about 7–10 years. This depends on the usage and maintenance, therefore, the ON an OFF operation of the generator is intended to maintain SOC within the 55% and 95% range, which ensures longevity of the system.

4.2.5. Price trend for the hybrid components

Information from US Energy Information Administration (EIA) and National Renewable Energy Laboratory (NREL) reports show that there has been a global decline in the price of PV module over the years. However, between the period 2005 and 2008 there was a spike in price due to silicon shortage in the market (Fig. 6). During the recession in 2009 and 2010 (resulting from the financial crisis of 2008) there was overcapacity issues as PV manufacturers over

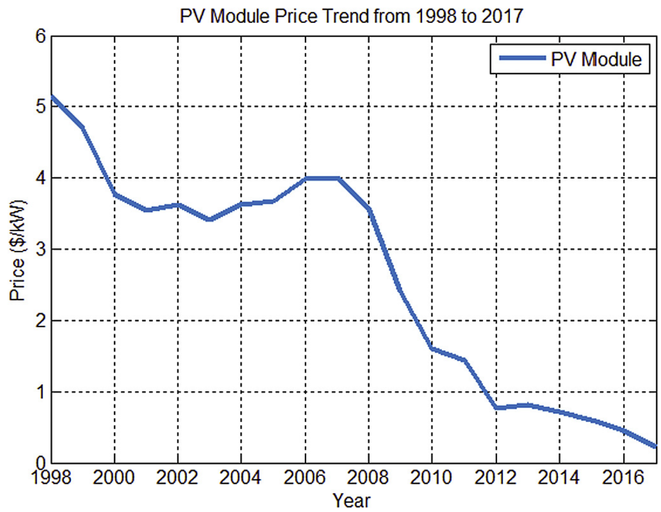

Fig. 6. PV price trend.

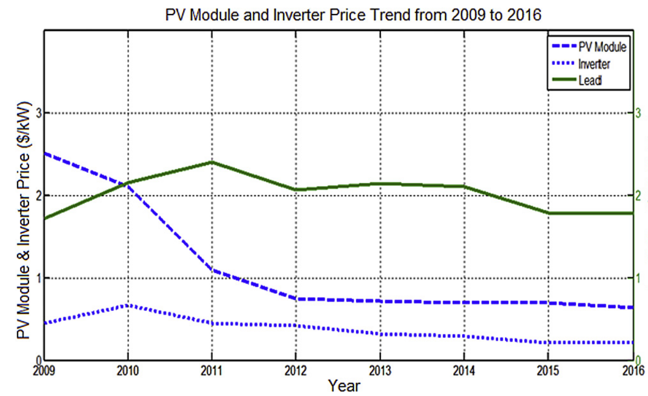

Fig. 8. Price trend from of renewable energy components from 2009 to 2016.

supply the market in order to keep up with demands, that in 2011 new additions outstrip the demand resulting in a sharp drop in the price of PV till date. Projections to 2018 suggests that the price will remain in the downward trend.

Diesel price has followed the global trend in crude oil and petroleum products market. The global financial crisis of 2008 also affected the price of diesel negatively during the period of 2008–2009 (Fig. 7). There was an increase in price between 2009 and 2010 and it remained relatively stable afterwards till 2014 before it begins to decline with no appreciable increase till date. In the same manner the price trend for inverters and lead spot price (which is a major determinant of the cost of lead acid battery) have been diminishing since 2009 (Fig. 8), [35–38].

4.2.6. Energy production by the hybrid mini-grid system

The data logger only measures the electricity supplied to the load, and thus does not measure the generated electricity. Nevertheless, simulation with HOMER™ indicates that the total energy produced by the hybrid mini-grid is 807,087 kWh/yr. Out of this the PV contributes 46% as renewable energy fraction (RF), while the rest is produced by the generator (Table 2). The total energy supplied to users is 643,367 kWh/year, representing about 80% of the total production. The 20% shortfall can be attributed to transmission losses and energy conversions in the inverters. The logged data from the master inverter indicates that the energy supplied to the load is 643,495 kWh/year, which is in close agreement with the simulated data.

Fig. 7. Diesel price trend.

4.2.7. Economics of operation of the hybrid mini-grid

Records from the plant operators show that the two 150 kW generators consume about 16 L of diesel per hour, while the 325 kW generator consumes about 32 L per hour. Calculations presented here are based on the two 150 kW generators, corresponding to a 300 KW Generator set in HOMER™ simulations. The average fuel consumption is 15.45 L per hour. The annual cost of diesel required to run the diesel generators is about US$150 000. The Net Present Cost (NPC) shows that the cost of acquisition, operation and maintenance of the generator constitutes about 64% of the total cost. Battery replacement and maintenance accounts for about 24% of the system cost, PV arrays and converters costs accounts for about 12% of the system cost (Table 3). The levelized cost of electricity production, obtained by dividing the total energy consumed (643,367 kWh/year) by the users by the total annualized cost (328,202), amounts to US$0.51 kWh, which corresponds to the value from the HOMER™ simulation (Table 4). The cost of electricity production was US$0.6/kWh with diesel only mini-grid. When the value of depreciation of Namibian Dollar to the US Dollar is taken into account the difference could be much bigger.

4.3. Contributing factors to the success of Tsumkwe hybrid mini-grid

Even though the sustainability of the Tsumkwe hybrid mini-grid is threatened by a number of challenges, it continues to deliver electricity to the village. A contributing factor to the survival of the hybrid mini-grid is the support from the government in the form of subsidies to cover the shortfall in revenue as well as providing skilled operators that have the capability to maintain the system. However, activities such as payments and replacement of spare parts are sometimes bogged down by institutional bureaucracy and bottlenecks. Replenishment of feedstock (diesel) is taken care of by the OTRC, which ensures that there are no shortages for the operation of the generators.

Another key factor that contributed to the success of the mini-grid system is that it was built into an existing system with paying customers, i.e. public and private institutions, and a community that has had a culture of paying for electricity. This most likely contributed to the high willingness to pay among the beneficiaries and therefore to a steady income from sales of electricity. The adoption of energy efficiency measures prior to the

Table 2
Energy production capacity of the hybrid mini-grid system.

Component	Capacity (kW)	Production (kWh/year)	Operational hour (h/year)	Electricity contribution (%)
PV	202	373,947	4343	46
Generator	300	433,140	3102	54
System		807,087		100

Table 3
Net present cost of electricity.

Component	ICC ($)	Fuel cost ($)	O&M ($)	Salvage ($)	Replacement ($)	Total cost ($)
PV	808,000	0	279,247	0	0	1,087,247
Generator	115,500	5,156,589	1,286,473	−173,729	826,250	7,211,083
Battery	275,760	0	330,915	118,915	2,215,537	2,703,297
Converter	150,000	0	67,393	90,611	214,298	341,080
System	1,349,260	5,156,589	1,964,028	−383,255	3,256,085	11,342,707

Table 4
Annualized cost of electricity.

Component	ICC ($)	Fuel cost ($)	O&M ($)	Salvage ($)	Replacement ($)	Total cost ($)
PV	23,379	0	8080	0	0	31,459
Generator	3342	149,206	37,224	−5027	23,908	208,653
Battery	7979	0	9575	−3441	64,106	78,219
Converter	4340	0	1950	−2622	6201	9869
System	39,041	149,206	56,829	−11,089	94,215	328,202

commissioning of the energy program reduced the load initially. However, this is now changing, the increasing population and influx of new inhabitants who were not privy to the existing energy efficiency measure, brought with them less energy efficient appliances to the village. The implementation of a progressive tariff has helped to ameliorate the burden of electricity on the low income households, and an effective prepaid metering system for electricity services has also facilitated easy revenue collection. The plant manager ascribed the success to the strict maintenance measures adopted for the operation and maintenance of the systems.

Another factor that contributes to the success of the mini-grid system is that the community was involved in the project from the beginning. This has resulted in community members still being involved in the operations and maintenance of the system, which is demonstrated by rendering services such as sanitary functions in the plant location on a voluntary basis.

5. Discussion

Access to sustainable energy is essential for a robust economy, sustainable ecosystems and the attainment of equity and social justice [40]. This report also argued that, lack of access to energy inhibits income generating activities and provision of basic services such as education and health care. Evaluation of the performances of technologies used for rural electrification programs in sub-Saharan Africa have shown that they are fraught with sustainability challenges as identified in Refs. [6,8,9,17,19and21]. These challenges were also manifest in Tsumkwe mini-grid program. However, the assessment of the program revealed why it has succeeded where others failed. Unlike most off-grid programs, Tsumkwe mini-grid has been operational for the majority of the time since it was commissioned in 2012, which is an achievement in comparison to the cases referred to above. Challenges such as lack of technical and managerial skills, disperse homesteads, low energy demand density, inappropriate financing models, lack of operational and managerial skills required to run and sustain projects after commissioning were addressed in Tsumkwe energy program. The technical personnel on ground have the requisite skills and experience to run and maintain the systems. Tsumkwe Village located at the center of Tsumkwe constituency has clustered homesteads with growing population and increasing energy demand as shown in the load profile trend. The challenges of high production are reduced by the inclusion of renewable components in the energy mix and discrepancy between revenue and cost were bridged through government subsidy. The assessment of the Tsumkwe hybrid mini-grid also suggest that the factors that have led to its success are:

- the involvement of the community members from the beginning of the project, this gave the users a chance to influence the development of the system. They also partake in voluntary services towards the upkeep of the systems.
- sensitization of the locals and creation of awareness on the importance of using energy efficient appliances to minimize costs.
- the use of a staggered tariff system, making the electricity affordable for poorer households, contributing to a general willingness to pay for electricity, which has been shown to be a problem at other locations [17, 41].

The advantage of this is evident when the financial situation of the diesel only mini-grid in Tsumkwe is brought into perspective. Information from OTRC office indicate that as of 2003 the end-user-tariff for the credit metering was about US$0.1/kWh, while institutions pay US$0.19/kWh. The cost reflective tariff then was about US$0.6/kWh necessitating the high tariff subsidy by OTRC local government [30]. This situation however left a deficit of about US$120,000 and a credit meter unpaid debt of about US$620,000 from residents who could not afford the bill [30]. These shortfalls have been addressed with the implementation of the prepaid metering system and good management of revenue ensuring financial sustainability. Feedstock (diesel) replenishment is handled by the OTRC, which ensures ring fencing of revenue for operation and maintenance services.

The decline in the cost of renewable energy renewable systems (Figs. 6–8) will encourage the addition of more PV, inverters and batteries in the network to increase the renewable energy fraction at a lesser cost. The analysis shows that presently, the cost of owning and operating the generator constitutes about 64% of the NPC, even though it represents only 9% of the ICC for the entire system. Therefore, increasing the renewable energy fraction will have positive impact on the financial stability and consequently sustainability of the program. Furthermore, the strict operation routine of the generators adopted by the manager, ensuring regular maintenance of the generators and the batteries, also, operation of the batteries within safe SOC, will contributes to sustainability of the system.

Nevertheless, there are several challenges that still threaten the sustainability of the Tsumkwe hybrid mini-grid. The analysis of data from the data logger indicates that the energy demand is increasing (Fig. 6). This is mainly caused by the increased home appliances and population, also there has been appreciable increase in social life. The awareness of energy efficiency measures is lower among the inhabitants that moved into the village after the commissioning of the project. This has resulted in an increasing number of electric stoves and electric heaters being used, which constitutes a challenge to the energy efficiency program adopted as a means of mitigating excessive energy use. There is an urgent need for a user awareness campaign directed at new comers to Tsumkwe, informing them about the need to use energy efficient appliances. This is necessary to reduce the cost of operation. However, the issue of none availability of spare parts is still a major source of concern, since acquisition of parts are delayed by bureaucratic bottlenecks.

Failed programs like the Lucingweni hybrid mini-grid pilot project in South Africa, Sekhutlane hybrid solar-biogas mini-grid in Botswana, and the Lesotho hydro mini-grids [17,20and21], lack various success elements associated with the Tsumkwe mini-grid. The Lucingweni program lacked the level of discipline noticed in Tsumkwe, unlike Tsumkwe the level of electricity pilferage through illegal connections was high, the willingness to pay for electricity was low; no metering system was in place and the 20 A fuse control system was ineffective [21, 41]. In addition, the program was introduced into a society that was not prepared. Sensitization through community engagement was not adequate and was handed over to an Independent Service Provider (ISP) without the necessary expertise to deal with the peculiarities of the local situation. A high tariff above the capacity and willingness of the people to pay was introduced in the middle of the program after providing free electricity for months [21, 41], which resulted in anger, vandalism and theft of the equipment. The Sekhutlane program had a strong focus on sensitization of beneficiaries at community meetings and development of business interests to increase payment capabilities among the locals [20]. Even so the project was unsuccessful due to technical issues, lack of financial responsibility and political reasons. Financial challenges also caused difficulties to the Lesotho hydro mini-grid program, the cost of production was high, and unlike Tsumkwe there was no leeway for the operators, the government did not bridge the gap in revenue, coupled with the fact that the four hydro plants suffered technical challenges due to siltation.

Tsumkwe hybrid mini-grid have shown that sustainability of rural electrification in sub-Saharan Africa is possible if challenges of operation and maintenance, financial transparency, and adoption of strict energy efficiency measures are addressed. The sustainability of Tsumkwe program has been mostly influenced by the government subsidy support and management, and the cooperation of the community throughout the establishment and implementation of the mini-grid system.

With the increasing population and businesses, a financial nexus may be attained in Tsumkwe village, where the revenue generated from electricity sales will be sufficient to cover the cost of electricity production without depending on subsidies from government. The lessons learnt from the assessment of the Tsumkwe hybrid mini-grid system could be replicated in sub-Saharan Africa and elsewhere in the developing world.

6. Conclusion

The objective of this study was to identify success factors and challenges from the functioning mini-grid installation in Tsumkwe, Namibia, to guide future implementation of such systems in rural off-grid communities in sub-Saharan Africa. The study concludes that:

- Even though, technology is a vital element of rural electrification initiatives, successful implementation and sustainability of these technologies require the involvement of the people and support from the government.
- The design of rural electrification systems should accommodate business interests, population growth and the corresponding load increases, to ensure long term sustainability of the systems.
- The adoption of energy efficiency measures is imperative for sustainable operations. Also, creation of awareness on energy efficiency measures should be an ongoing thing to forestall the possibility of unaware new customers using less efficient appliances.
- A progressive tariff system is essential to accommodate low income households in the energy matrix.
- Capacity development of the locals in technical skills, management and control could lead a more sustainable program.
- Prepaid metering system is more effective for revenue collection and in curtailing electricity theft in rural electrification programs.
- It is possible for Tsumkwe mini-grid experience to be extrapolated in other sub-Saharan African countries, if the challenges of limited technical and managerial stills, availability of spare parts, uninterrupted supply of feedstock, lack of fiscal responsibility and a genuine transfer of technology through capacity building of locals is properly addressed.

It is recommended that the design of rural electrification programs should be scalable to cater for increasing population and businesses. Stocking of spare parts and feedstocks could reduce power downtime during shortfalls and contingencies and thus increase sustainability of operation. Initially, management and control of energy programs is best handled by the government who have the financial and political resources to handle the oddities of off-grid electrification programs, before handing over to an independent service provider (ISP).

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